ERICK OLUGA ABOGE

Full Stack Software Engineer | React · TypeScript · Node.js · Python | AWS, Terraform & CI/CD | AI-powered product engineering

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PROFESSIONAL SUMMARY

Full stack engineer with 5+ years building software, four of them taking applications from ambiguous brief to live production — architecture, front end, APIs, data layer and the infrastructure underneath. Expert in React/TypeScript and Node.js on PostgreSQL, deployed on AWS with Terraform, Docker and gated CI/CD, with a second cloud footprint on Google Cloud Run. Build AI-powered product features rather than only using AI to code: lead engineer on an AI spend intelligence product line covering provider usage ingestion, invoice-level reconciliation, a 13-detector anomaly engine and automated audit reporting, plus a published MCP server exposing live payment infrastructure to LLM clients through tool use. Comfortable owning ambiguity end to end — co-founded a multi-tenant SaaS platform from market research through AWS production deployment, and delivered 15+ projects for international clients across e-commerce, finance and media. Deliberate about where AI accelerates work and where human judgement has to take over: every generated line is validated against security and business constraints before it ships.

TECHNICAL SKILLS

Frontend: React · Next.js · SvelteKit · TypeScript · JavaScript (ES6+) · Tailwind CSS · Redux · Zustand · Context API · Framer Motion · Component architecture · Highly interactive and real-time UIs

Backend: Node.js · Express · Fastify · Python (Django · FastAPI) · Go · REST API design · OpenAPI · Microservices · Webhooks · JWT · OAuth2 · RBAC · C# / .NET Core

Cloud & DevOps: AWS (Lambda · SQS · SNS · S3 · RDS · App Runner · Amplify · SES · IAM) · Google Cloud Run · Azure (familiar) · Terraform (IaC) · Docker · GitHub Actions CI/CD with gated production deploys · Linux · Git

Data & Infrastructure: PostgreSQL · Redis · MySQL · SQL Server · Schema design · Row-Level Security · Indexing and query optimisation · Migrations · Serverless functions · API gateways · Prisma · Drizzle · Django ORM

Data Pipelines: Ingestion and ETL across third-party providers · Inbound email pipelines · Content-hash deduplication · Event-driven async workers and queues · Inngest · Billing reconciliation and anomaly detection

AI Engineering: Anthropic / Claude API · OpenAI · Tool use and function calling · Prompt design · MCP and FastMCP servers · AI agents · RAG pipelines · LangChain · Evaluating model output in production · Claude Code · GitHub Copilot

Quality & Testing: Unit · Integration · End-to-end · Vitest · pytest · Jest · TDD · Automated test gates in CI · Structured logging · Grafana · Loki · Metrics and production debugging

Practices: Clean architecture · Domain-driven design · Modular and reusable component design · Multi-tenancy · Agile/SCRUM · Code review · Technical design documentation · Stakeholder and requirements partnering

PROFESSIONAL EXPERIENCE

Senior Software Engineer — Dash.fi

Austin, Texas, United States · Remote | Contract | June 2026 – Present

- Own features end to end across the fintech core — card transaction surfaces and admin tooling, AP draft-bill automation, approval workflows and accounting integrations (QuickBooks/NetSuite via Rutter) — from design through API, front end, tests, deployment and production accountability.
- Lead engineer on the AI Spend Agent, the company's newest product line: provider usage and cost ingestion, invoice-level billing reconciliation, a 13-detector anomaly engine, budget-breach alerting (Inngest with Knock/SendGrid and once-per-period guarantees) and server-side PDF audit reports — an AI-powered product, not an AI-assisted workflow.
- Built invoice ingestion infrastructure for bill pay: inbound email pipelines (SES forwarding, webhook-driven mailbox sync) that classify billing documents by sender and route them with content-hash deduplication.
- Integrated banking and risk data vendors — Plaid balances, Middesk business verification, Heron financial enrichment — behind response-shape guards so vendor degradation never breaks a customer request; implemented RBAC and user lifecycle on Stytch B2B with per-route enforcement at the API gateway.
- Own my services' infrastructure as code: Terraform across Lambda, SQS, S3, IAM and PostgreSQL roles, with CI/CD including gated production deploys and Grafana/Loki observability.

Senior Software Engineer — Umba

Nairobi County, Kenya · Remote | Full-time | June 2026 – Present

- Part of Umba Group's engineering organisation, currently assigned to the Dash.fi engineering team building global fintech products.

Co-Founder & Lead Engineer — Carbon Corp Solutions Limited

Nairobi, Kenya | September 2024 – Present

- Took a multi-tenant enterprise SaaS platform from ambiguous market opportunity to live production — market research, architecture, implementation, test strategy and AWS deployment — the full concept-to-production arc with no brief handed down.
- Designed a 41-table PostgreSQL schema with Row-Level Security enforcing tenant isolation at the database level, plus a dual-pool connection strategy using AsyncLocalStorage; published the architecture publicly as a technical write-up.
- Built a reusable configuration-driven form and workflow layer — 150+ KPI data entry forms, a two-pass calculation engine and a two-level approval workflow with separation-of-duties — designed so new tenants and sectors are configured rather than rebuilt.
- Ran Docker containerisation, path-filtered GitHub Actions and unit/integration suites across AWS (App Runner, Amplify, RDS, S3, SES), with automated PDF board reporting on the front end.

Full Stack Engineer — Optiven Limited

Nairobi, Kenya · Hybrid | August 2024 – May 2026 · converted from contract to full-time on delivery quality, December 2024

- Owned the full development lifecycle for internal platforms — requirements gathering with non-technical stakeholders through to production deployment — across an HRMS performance module (8-state approval workflow serving 150+ employees), a workforce compliance system and a CRM.
- Architected a CRM over 10,000+ customer records in Django and React/TypeScript; composite indexing, query tuning and server-side pagination cut query time by 40%.
- Designed and deployed a production M-Pesa payment integration — idempotent webhook handlers keyed on transaction references, explicit timeouts and multi-state transaction tracking (pending/confirmed/failed/reversed) with automated reconciliation — reducing reconciliation errors by 90%.
- Integrated Microsoft Dynamics 365 through C#/.NET Core APIs to surface real-time financial data on executive dashboards without disrupting the system of record; containerised services with Docker and owned GitHub Actions CI/CD with automated test gates in a 5-person Agile team.

Software Engineer (Contract) — GMC Kitengela

Nairobi County, Kenya · Remote | January – August 2024

- Engineered a procurement module for a point-of-sale system (React + Django), digitising an undefined manual process into a structured, role-gated workflow: Requisition → RFQ → Bid Analysis → Purchase Order → GRN → Invoice. Improved inventory accuracy and cut approval turnaround.

Full-Stack Software Developer — Upwork (Freelance)

Remote | November 2022 – January 2024

- Delivered 15+ projects with a 5-star rating across e-commerce, finance and news platforms for US and European clients — each a different domain, stack footprint and stakeholder, scoped and delivered independently with React, Node.js and .NET services including payment gateway integrations.

SELECTED PROJECTS — AI ENGINEERING & SYSTEMS

M-Pesa Express MCP Server · Python · FastMCP · Google Cloud Run

- Built and deployed an MCP server exposing live M-Pesa Daraja payment infrastructure to LLM clients through tool use, so payments can be orchestrated in natural language — tool schema design, function calling and deployment on a second cloud platform.

Ask the Web — AI Research Agent · Python · LangChain · RAG · Streamlit · Groq · Ollama

- Production RAG pipeline: query understanding, tool orchestration, web retrieval and cited synthesis, with model output evaluated against source rather than trusted. GitHub

MiniDB — Relational Database Engine from Scratch · Python · pytest · B-tree indexing

- SQL lexer, recursive descent parser, query planner, executor and B-tree index built from scratch with 49 automated tests. Live · GitHub

Nexus Financial · Next.js · TypeScript · Prisma · AI categorisation

- Statement parsing with AI transaction categorisation and interactive analytics dashboards. Live · GitHub · Carbon Corp (live)

EDUCATION

Bachelor of Commerce — Finance (Second Class Upper), University of Nairobi

Certificate in Software Development, Moringa School

ADDITIONAL

Languages: English (fluent) · Swahili (native) **Location:** Nairobi, Kenya · Kenyan citizen with valid Kenyan work authorization

Writing: Technical write-ups on multi-tenant isolation in PostgreSQL and system architecture